



Birzeit University

Faculty of Engineering and Technology

Department of Electrical and Computer Engineering

First Semester – 2023/2024

ENCS4320 - Applied Cryptography

Homework # 1 - Due Monday, May 13, 2024

Instructions: You must show your steps in the area below every question, if any. Not showing your work for any question (if work is required) results in a score of ZERO.

Question 1 (6 points):

Using your cryptanalysis skills, find the *plaintext* (and the *key*) that corresponds to the ciphertext **WLIMWXLIWSYPSJQCWSYP**, given that the *shift cipher (ROT-k)* was used.

Question 2 (11 points):

Assume an attacker knows that a user's password is either "**wxyz**" or "**bddf**". Say the user encrypts his password using:

- a) The substitution cipher,
- b) The Vigenère cipher using period 2, or
- c) The Vigenère cipher using period 3

and the attacker sees the resulting ciphertext. Show how the attacker can determine the user's password, or explain why this is not possible.

Question 3 (6 points):

Suppose we have a computer with a 4.2 GHz 16-core processor that executes 4.2×10^9 cycles per second per core. Considering that it can test 20 keys per CPU cycle, calculate:

- a) The average time (in years) to find a key by the brute-force attack if the key size is **56** bits.
- b) The average time (in years) to find a key by the brute-force attack if the key size is **128** bits

Question 4 (16 points):

Answer each of the following without using a calculator.

- a) $3 - 11 \pmod{9} =$
- b) $15 \times 29 \pmod{13} =$
- c) $-12 / 35 \pmod{19} =$
- d) Are 172 and 68 co-prime numbers?

Question 5 (8 points):

Consider a cryptosystem that encrypts a message $m \in \mathcal{M}$, $\mathcal{M} = \{0, 1, 2\}$, using a shared key k that is uniformly chosen from the key space $k \in \mathcal{K}$, $\mathcal{K} = \{0, 1, 2, 3\}$.

$$c = \text{Enc}(k, m) = m + k \pmod{3}, \text{ and}$$

$$m = \text{Dec}(k, c) = c - k \pmod{3}$$

Is this encryption scheme *perfectly secret*? Explain your answer.

Question 6 (8 points):

The following question concerns multiple encryptions of single-character ASCII plaintexts with the **one-time pad** using the same 8-bit key. You may assume that the plaintexts are either (upper-case or lower-case) English letters or digit characters (0, 1, ..., 9). Say you see the three ciphertexts $(\text{EF})_{16}$, $(\text{A4})_{16}$, and $(\text{D3})_{16}$. What can you deduce about the plaintext characters these correspond to?

Bit 4:	0000000000000000	1111111111111111	
Bits 0-3:	0123456789ABCDEF	0123456789ABCDEF	Block:

Bits 00			Control characters
5-6: 01	!"#\$%&'()*+,-./	0123456789;<=>?	Numbers and punctuation
10	@ABCDEFGHIJKLMNO	PQRSTUVWXYZ[\]^_	Uppercase letters (mostly)
11	`abcdefghijklmno	pqrstuvwxyz{ }~.	Lowercase letters (mostly)

Question 7 (10 points):

Suppose that, after a particular step of **A5/1 stream cipher**, the values in the registers are:

$$X = (x_0, x_1, \dots, x_{18}) = (10101010101010110)$$

$$Y = (y_0, y_1, \dots, y_{21}) = (1100110001101100010011)$$

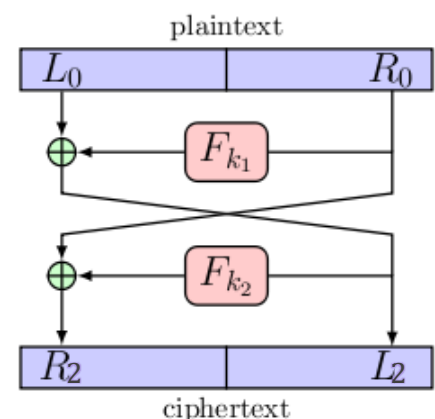
$$Z = (z_0, z_1, \dots, z_{22}) = (11100101110000011000011)$$

- List the next 4 keystream bits.
- Give the contents of X, Y, and Z after the generation of each of these 4 bits.

Question 8 (15 points):

Consider a new **block cipher**, **DES2**, that consists of only *two rounds* of the **DES** block cipher. **DES2** has the *same block and key size* as DES. For this question, you should consider the DES **F** function as a black box that takes two inputs, a 32-bit data segment, and a 48-bit round key, and produces a 32-bit output. Using the chosen-plaintext attack (CPA) without any restrictions on the number of oracle calls.

- Give an algorithm to recover the 48-bit round keys for round 1 (k_1) and round 2 (k_2). Your algorithm should have fewer operations than the exhaustive key search for **DES2**.
- Can your algorithm be converted into a distinguishing attack against **DES2**, i.e., an attack that distinguishes **DES2** from a random permutation?



Question 9 (20 points):

This problem deals with the *AES-128 block cipher*.

- a) Assume that the first column of the input to the **InvMixColumn** step is $S_{i,0} = (B4, 52, E0, AE)_{16}$, find the 3rd element of the corresponding column of the output state of the **InvMixColumn** step.
- b) Given the input $S_{2,1} = (7A)_{16}$, find **InvSubByte**($S_{2,1}$).
- c) Assume that round key 6 (k_6) is $(98\ 0F\ 71\ AF\ 15\ C9\ 47\ D9\ 0C\ B7\ E8\ 59\ D6\ 7F\ 67\ AD)_{16}$, find the round keys for round 5 (k_5) and round 7 (k_7).

GOOD LUCK